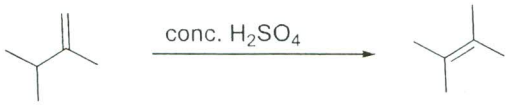
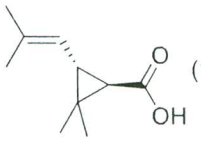
**Organic Cumulative Examination**
January 20, 2017

1) Explain how IR spectroscopy might be used to monitor the progress of the following reaction (**10 points**)

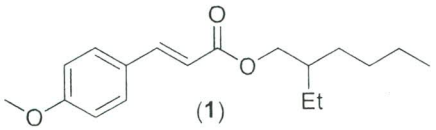



2) Chrysanthemic acid (**1**) is isolated from chrysanthemum flowers. The IR spectrum of (**1**) exhibits five signals above 1500cm$^{-1}$. Identify the course for each of these signals. (**10 points**)




(**1**)

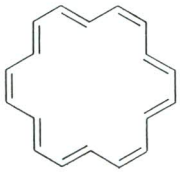
3) You work in an international company that develops sunscreen lotions based on organic compounds. A typical example is compound (**1**)




(**1**)

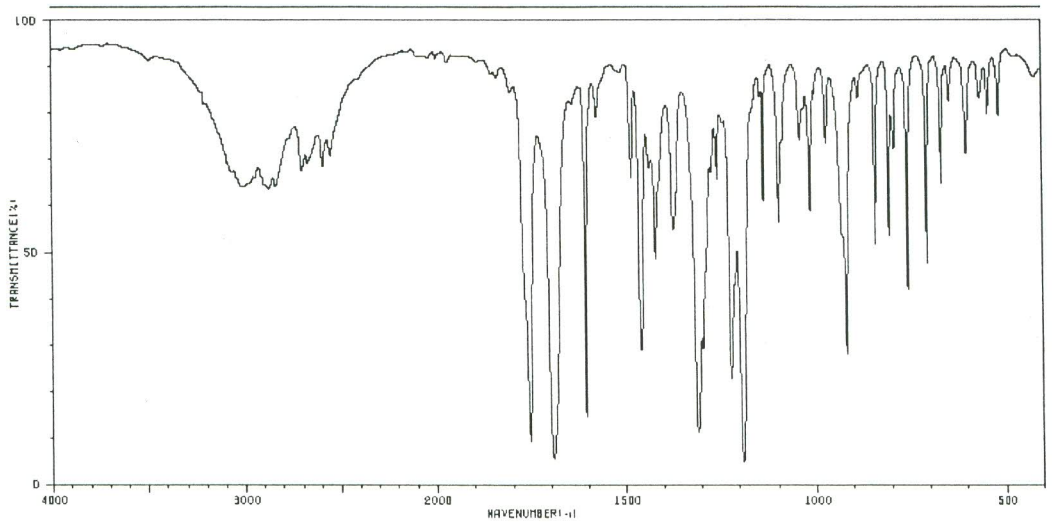
a) Discuss the potential of the compound to protect against UV-A and/or UV-B light (UVA: 315-400nm, UVB: 280-325nm) (**10 points**)
b) Your technician suggests to make a derivative of the compound as new lead compound to bring in millions of dollars, and use an unsaturated alcohol for the needed ester synthesis. Would it be beneficial for the functionality of the sunlotion to actually prepare such a compound? Give a rational to support your statement. (**20 points**)

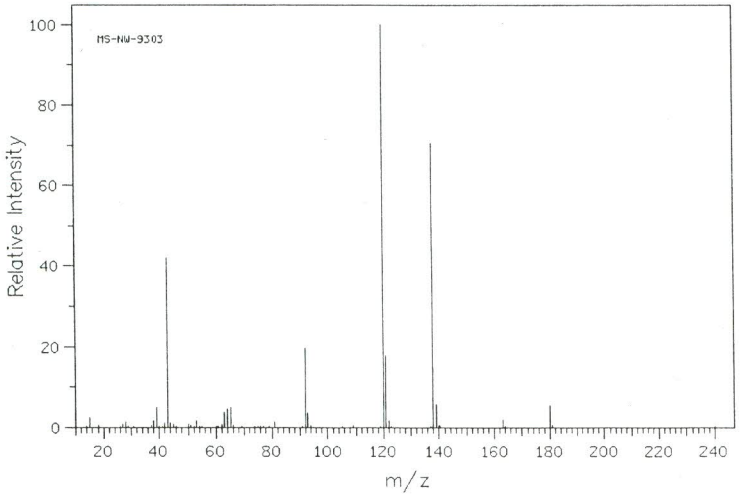
4) Cyclooctadecanonaene, [18]annulene, has been described in JACS, 1962 for the first time by Sondheimer et al from the UK. The reddish-brown compound is relatively stable at ambient temperature. Computational chemists, however, questioned the claimed aromaticity based on their calculations.
**Which experiment(s)** would you perform to get more information on whether or not this compound behaves as an aromatic compound versus a polyolefin? Discuss your suggestions. (**25 points**)

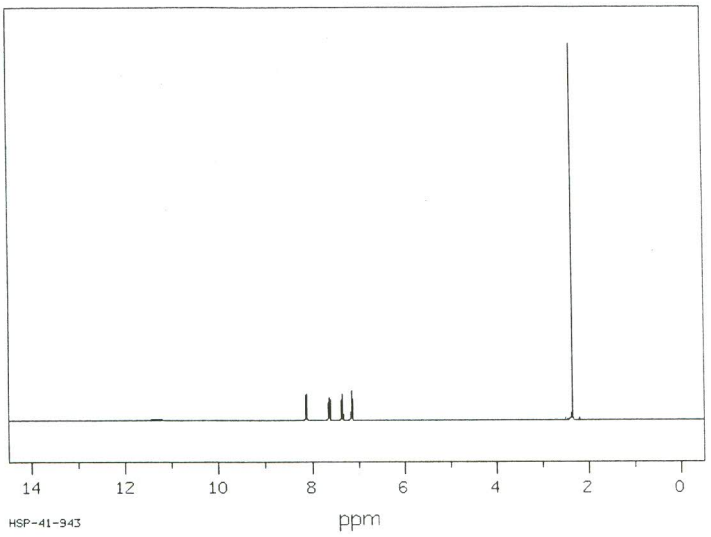


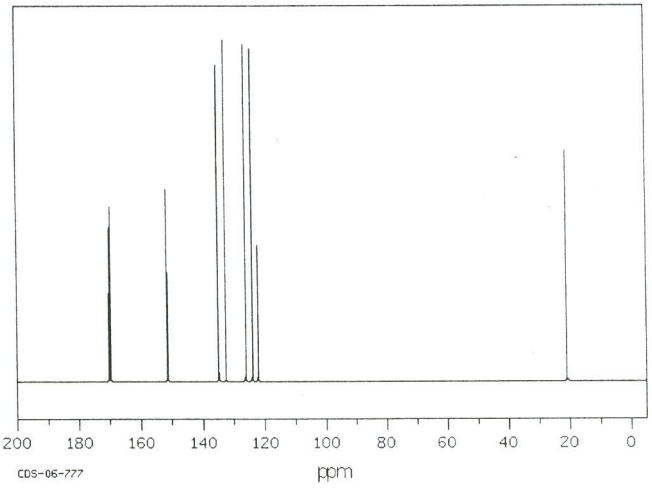
5) The following spectra were recorded from an unknown (?) substance $\mathbf{C_9H_8O_4}$. Make a reasonable suggestion for its structure, explain how you concluded that and name the compound! (**25 points**)

Hint(s): The $^1$H NMR spectrum has signals that integrate to 1:1:1:1:1:3. When measuring the $^1$H NMR spectrum in MeOD-$d_4$ or $D_2O$, the tiny signal at 11.2ppm disappears. I hope you do not get a headache here...




14
12
10
8
6
4
2
0
HSP-41-943
ppm


200
180
160
140
120
100
80
60
40
20
0
CDS-06-777
ppm